

EE 5239 Nonlinear Optimization

Lecture 2: Gradient Methods I: Introduction

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Outline

- 1 Review
- 2 Optimization Methods: Motivation
- 3 Gradient Descent Method

Announcement

- First HW due Sept 20th night (Friday).
- Note the requirement for typing the HWs.
- Today, cover Bertekas Sec. 1.2. Gradient Methods.

Last Time

$$\begin{array}{ll}\text{minimize} & f(x) \\ \text{subject to} & x \in \mathbb{R}^K\end{array}$$

- Studied the unconstrained optimization problem
- Characterized the local optimal solution of this problem
- When does local min becomes global min?

Last Time

- Necessary condition

$$\begin{aligned}\nabla f(x^*) &= 0, && \text{(first-order condition),} \\ \nabla^2 f(x^*) &\succeq 0, && \text{(second-order condition).}\end{aligned}$$

- Sufficient condition

$$\begin{aligned}\nabla f(x^*) &= 0, && \text{(first-order condition),} \\ \nabla^2 f(x^*) &\succ 0, && \text{(second-order condition).}\end{aligned}$$

- How to use optimality condition to directly find the (local) optimal solutions

Last Time

- We will call x^* that satisfies the first-order optimality condition

$$\nabla f(x^*) = 0, \quad (\text{first-order condition}),$$

as a stationary point

- Exercise: Quadratic problems?

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Motivation

- How to find a stationary solution?
- **Easiest way:** Simply solve $\nabla f(\mathbf{x}) = 0$!
- Suppose $\nabla f(\mathbf{x}) = 0$ is not easy, don't know how to solve
- Then what?

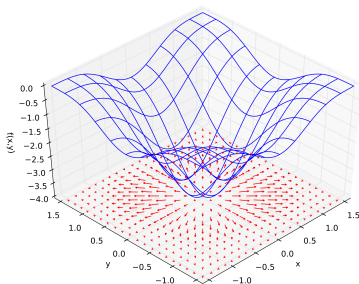


Figure 0.1: The gradients of a function (Wikipedia: Gradient)

Example

- The well-known logistic regression problem is given below

$$\min L(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N \ln \left(1 + e^{-b_i \mathbf{x}^T \mathbf{a}_i} \right)$$

- Logistic regression has wide applications, estimating probability, classification, etc
- **Question:** Is this problem “convex”? How to find its solution?

Example

- First attempt, finding the gradient

$$\nabla L(\mathbf{x}) = -\frac{1}{N} \sum_{i=1}^N \frac{b_i \mathbf{a}_i}{1 + e^{b_i \mathbf{x}^T \mathbf{a}_i}}$$

- Setting it to zero, closed-form solution for $\mathbf{x}$?

Motivation

- **Main Strategy:** Test a series of points, hopefully arrive at a point $\nabla f(\mathbf{x}^*) = 0$
- How to test? Design an algorithm!
- **Intuition:**
 - ① The gradient of f points to a direction that **increases** the objective
 - ② I want my next test point to be “**opposite**” with the gradient direction
- Consider the problem $\min f(x) = x^2$

Two strategies for algorithm design

- Line search methods
- Trust region methods

[discussion on board]

The Gradient Descent Method

- We will start with a family of classical method: [Gradient Descent](#)

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- If $\nabla f(\mathbf{x}) = 0$, then $\mathbf{x}$ is a candidate solution (satisfying first-order sufficient condition); Done

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- If $\nabla f(\mathbf{x}) \neq 0$, there is an interval $(0, \delta)$ of stepsizes such that

$$f(\mathbf{x} - \alpha \nabla f(\mathbf{x})) < f(\mathbf{x}), \forall \alpha \in (0, \delta).$$

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- Show this using **Mean Value Theorem** [on board]?

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- Show this using **Mean Value Theorem** [on board]?
- More generally, if a given direction $\mathbf{d}$ that is with **obtuse angle** with $\nabla f(\mathbf{x})$

$$\langle \nabla f(\mathbf{x}), \mathbf{d} \rangle < 0$$

there is an interval $(0, \delta)$ of stepsizes such that [try to prove]

$$f(\mathbf{x} + \alpha \mathbf{d}) < f(\mathbf{x}), \forall \alpha \in (0, \delta).$$

Iterative Descent Methods

$$\mathbf{x}^{r+1} = \mathbf{x}^r + \alpha_r \mathbf{d}^r, \quad r = 0, 1, \dots$$

where, if $\nabla f(\mathbf{x}^r) \neq 0$, the direction $\mathbf{d}^r$ satisfies $\nabla f(\mathbf{x}^r) \mathbf{d}^r < 0$, and α^r is a positive stepsize

- **General Case:** Gradient descent methods

$$\mathbf{x}^{r+1} = \mathbf{x}^r - \alpha_r \mathbf{D}^r \nabla f(\mathbf{x}^r), \quad r = 0, 1, \dots$$

where $\mathbf{D}^r$ is a positive definite matrix called **scaling matrix**

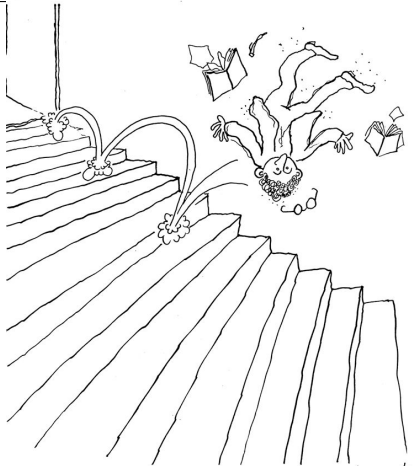
- **Special case I:** Steepest descent

$$\mathbf{x}^{r+1} = \mathbf{x}^r - \alpha_r \nabla f(\mathbf{x}^r), \quad r = 0, 1, \dots$$

- **Special case II:** Newton's method

$$\mathbf{x}^{r+1} = \mathbf{x}^r - \alpha_r \left(\nabla^2 f(\mathbf{x}^r) \right)^{-1} \nabla f(\mathbf{x}^r), \quad r = 0, 1, \dots$$

Discussion



*Just after learning the "Steepest Descent" method
in optimization class...*

Figure 0.1: The Steepest Descent (ERASIP: DSPHumour)

Discussion

- In practice steepest descent may have slow convergence
 - Practical performance?
 - **Exercise:** implement the steepest descent for a 2-D convex quadratic problem. Show the convergence plot on the contour of the function

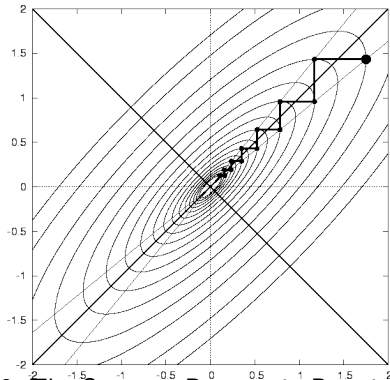


Figure 0.2: The Steepest Descent in Practice (Komarix.org)

Discussion

- Newton's method: generally fast convergence
 - ① It treats the objective (locally) as a quadratic problem around $\mathbf{x}^r$

$$f(\mathbf{x}) \approx f(\mathbf{x}^r) + \langle \nabla f(\mathbf{x}^r), \mathbf{x} - \mathbf{x}^r \rangle + \frac{1}{2}(\mathbf{x} - \mathbf{x}^r)^T \nabla^2 f(\mathbf{x}^r)(\mathbf{x} - \mathbf{x}^r)$$

- ② So solving the right hand side, we obtain

$$\mathbf{x}^{r+1} = \mathbf{x}^r - (\nabla^2 f(\mathbf{x}^r))^{-1} \nabla f(\mathbf{x}^r)$$

- ③ **Question:** how many iterations does it take for Newton method to minimize a quadratic function f ?
 - ④ **Caution:** very difficult to make it numerically stable, needs more information than the steepest descent method

Choice of Stepsize

- **Constant Stepsize:**

$$\alpha_r = \alpha$$

Comment: practically used often, but what's the constant?

- **Minimization Rule:** Pick α_r such that

$$\alpha_r = \arg \min_{\alpha \geq 0} f(\mathbf{x}^r + \alpha \mathbf{d}^r)$$

Comment: maximum reduction, but solving the optimization problem may be expensive

- **Limited Minimization Rule:** Pick α_r such that

$$\alpha_r = \arg \min_{\alpha \in [0, s]} f(\mathbf{x}^r + \alpha \mathbf{d}^r)$$

Choice of Stepsize (Cont.)

- **Diminishing Stepsize:** useful in practice, but cannot diminish too fast? Why??

$$\alpha_r \rightarrow 0, \quad \sum_{r=1}^{\infty} \alpha_r = \infty$$

- **Armijo rule:** Let $\sigma \in (0, \frac{1}{2})$. Fix s as a constant, and $0 < \beta < 1$ as a constant, Keep shrinking α by $s, \beta s, \beta^2 s, \dots$ until the following is satisfied

$$f(\mathbf{x}^r + \alpha \mathbf{d}^r) - f(\mathbf{x}^r) \leq \sigma \alpha \langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle$$

Comment: achieves sufficient descent, may need to test many times.

The Armijo Rule

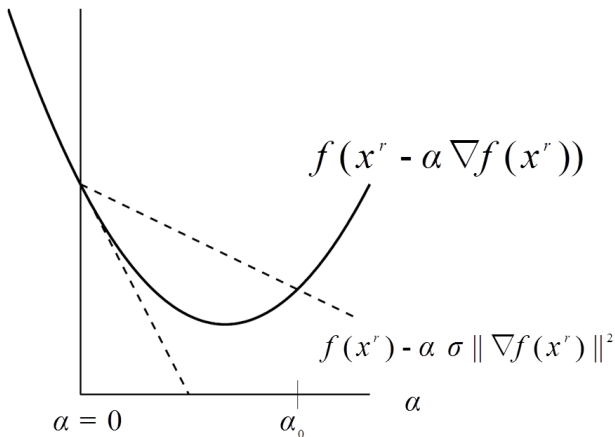


Figure 0.3: The Armijo Stepsize Selection

The Overall Strategy

- No matter what strategy we choose, there should be **sufficient descent** in the objective at each step
- The objective function $f(\mathbf{x})$ serves as a “potential” to guide the optimization process
- These methods are called “**descent**” methods, for precisely this reason
- Basically a “good” stepsize and a “good” direction is all that is required to find the (local) optimal solutions
- Next time: theoretical analysis of descent methods